

Who's who

All the scientific knowledge we have today is the result of centuries of careful questioning, research, and observation by the world's most brilliant

minds. Listed below are some of the most inspiring scientists, inventors, mathematicians, and philosophers throughout history.



Alhazen (965–1040)

Arab mathematician, astronomer, and physicist who made a significant contribution to the field of optics (the study of light and vision). He devised the laws of reflection and refraction, and described the anatomy of the human eye.

Al-Khwarizmi (c 780–c 850)

Persian mathematician, geographer, and astronomer widely known as the “Father of Algebra”. He was responsible for introducing Arabic numbers to Europe. Working in Baghdad (in present-day Iraq), he produced two mathematical textbooks and important works on geography and astronomy.

Al-Razi (c 854–c 925)

See p.35

Anning, Mary (1799–1847)

See p.115



Archimedes (c 287–c 212 BCE)

Greek inventor, philosopher, and mathematician born in Syracuse on the east coast of Sicily in Italy. He is known for discovering the principles of flotation and invented the Archimedes Screw. He also constructed siege machines to defend Syracuse against the Romans.

Aristotle (384–322 BCE)

See pp.30–31

Babbage, Charles (1791–1871)

British mathematician and inventor, often referred to as the “Father of Computing”. He spent his working life building two mechanical computers that could store information. Although he never completed the machines, they are regarded as the forerunners of the modern computer.

Bacon, Roger (c 1214–1292)

See pp.48–49



Baird, John Logie (1888–1946)

Scottish engineer, inventor, and television pioneer. Baird succeeded in transmitting the outline of shapes in 1924 and moving objects in 1926. He produced the first color television pictures in 1928.

Barnard, Christiaan Neethling (1922–2001)

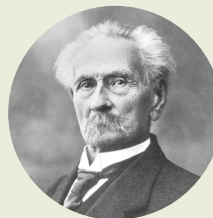
South African surgeon who was a pioneer of open heart surgery and performed the first successful human-to-human heart transplant in 1967. His patient, a grocer called Louis Washkansky, received the heart of a car accident victim, but died from pneumonia 18 days later.

Bassi, Laura (1711–1778)

See p.98

Bell, Alexander Graham (1847–1922)

See p.150



Benz, Karl (1844–1929)

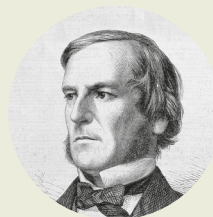
German engineer and car manufacturer. Working with Gottlieb Daimler, he built the first successful internal combustion motor car in 1885. In 1893, he produced the first four-wheel motor vehicle. The Benz company began producing the world's first race cars in 1899.

Biot, Jean-Baptiste (1774–1862)

French physicist, astronomer, and mathematician who established the existence of meteorites and developed a technique for analyzing sugar solutions. In 1804, he was one of the scientists on board the first scientific balloon flight. Working with fellow physicist Félix Savart, he demonstrated a connection between electricity and magnetism in 1820.

Bohr, Niels (1885–1962)

See p.168



Boole, George (1815–1864)

British mathematician whose work on logic laid many of the foundations for modern computer science. He devised a system of logic known as Boolean logic—a form of algebra that is basic to the design of modern digital computer circuits.

Bosch, Carl (1874–1940)

German industrial chemist who won the 1931 Nobel Prize in Chemistry. He developed a process called the Haber–Bosch process that combined hydrogen and nitrogen to produce ammonia. This process made it possible to produce enormous quantities of fertilizers and explosives.

Brahe, Tycho (1546–1601)

See p.67